Resilient Federated Decision Intelligence for UAV Swarm Target Tracking and Coordination Across Heterogeneous Mission Environments

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***Abstract*—This paper introduces a resilient federated decision intelligence system for the tracking and coordination of autonomous UAV swarm targets in heterogeneous mission scenarios. In the conventional centralized swarm control system, communication bottlenecks and single points of failure are inherent, which are not scalable and resilient in dynamic operational settings. To overcome these limitations, we introduce a decentralized multi-agent reinforcement learning system where each UAV learns a local Proximal Policy Optimization (PPO) policy and periodically engages in communication-aware federated aggregation. A leader-based aggregation strategy is used to make the system fault-tolerant and communication dropout-resilient, with adaptive hierarchical formation reconfiguration for self-healing swarms. The system is designed to be adaptable to different environments such as defense surveillance, agricultural monitoring, underwater exploration, and wildfire detection with adaptive reward modeling. Experimental results are planned to be conducted on the AirSim simulator with a five-UAV scenario setup, which can be scaled up to larger swarms.**

***Index Terms*—Federated Learning, UAV Swarm, Multi-Agent Reinforcement Learning, PPO, Resilient Coordination, Heterogeneous Environments.**

I. Introduction

Unmanned aerial vehicle (UAV) swarms are being identified as a crucial component of mission-centric applications such as defense surveillance, environmental monitoring, agricultural inspection, disaster relief, and maritime exploration [1], [2]. In traditional centralized swarm systems, coordination is done through ground control stations or cloud servers. Centralized systems are efficient for small-scale swarms, but they are susceptible to communication bottlenecks, scalability problems, and single points of failure [3], [4].

Recent advances in multi-agent reinforcement learning (MARL) have made it feasible to implement decentralized decision-making, where control policies are learned autonomously by each UAV [5]. However, autonomous learning results in inefficient coordination and conflicting policies among agents [6]. Federated learning (FL) is a more effective approach for collaborative learning with distributed model aggregation, without exchanging raw data [7], [8]. This paper presents a robust federated decision intelligence framework for autonomous target tracking and coordination of UAV swarms in heterogeneous mission environments. The framework combines local Proximal Policy Optimization (PPO)-based learning with communication-aware federated aggregation and leader election. The proposed system improves communication dropout and agent failure robustness while preserving decentralized execution. The proposed framework is flexible for heterogeneous mission environments such as defense surveillance, wildfire tracking, agricultural evaluation, and maritime surface observation missions involving distributed perimeter search and return actions [9], [10].

The key contributions of this research work are:

- A communication-aware federated multi-agent reinforcement learning framework for decentralized UAV swarms.
- A rotating leader-based aggregation scheme for fault-tolerant policy updates.
- A hierarchical formation reconfiguration strategy for failure handling.
- An environment-adaptive reward design for heterogeneous missions.
- An AirSim-based validation framework for resilient swarm coordination.

II. Related works

The research on the intelligence of UAV swarms has moved from a centralized approach towards decentralized and federated learning-based approaches. In this regard, the following section discusses the latest research on federated learning, federated reinforcement learning, and fault-tolerant coordination in the context of UAV swarms.

A. Federated Learning in UAV Swarms

Several studies have been conducted to investigate hierarchical federated learning with the help of UAV swarms to speed up convergence. Energy-efficient decentralized federated learning with leader election and spiking neural networks has been researched to reduce communication overhead. In clustering-based federated learning, UAVs are grouped depending on data similarity or optimization directions, which

TABLE I: Comparison of the Proposed Framework with Existing UAV Swarm FL/RL Approaches

Approach / Work	Cent.	Rot.	Drop.	Adapt.	Hier.	PPO	AirSim	Res.
Hierarchical FL/RL [9], [11]	✓	✗	✗	✓	✗	✗	✗	Low
Decentralized FL + Leader [3], [7]	✗	✓	✗	✗	✗	✗	✗	Medium
Clustered FL in UAV [6], [13]	✓	✗	✓	✗	✗	✗	✗	Medium
Asynchronous FL [29]	✗	✗	✓	✗	✗	✗	✗	Medium
FedMARL for UAV [10], [14], [17], [21], [24]	✗	✗	✓	✗	✓	✓	✓	Low–Medium
Proposed Framework (Ours)	✗	✓	✓	✓	✓	✓	✓	**High**

is helpful when data is distributed heterogeneously among devices [13], [20]. Resource allocation and joint training strategies were researched to make federated learning more realistic for UAV networks [11].

B. Reinforcement Learning and Federated RL for Swarm Coordination

Researchers have compared centralized and federated reinforcement learning methods for swarm coordination, finding that federated variants often show better transfer performance across environments but remain sensitive to communication interruptions [14]. Asynchronous federated RL has been proposed to cope with unreliable links by applying data-value-based weighting [29]. Multi-task federated RL frameworks introduce dynamic knowledge sharing to handle varying task demands among UAVs [12]. In parallel, multi-agent RL studies for UAV control outline methods for coordination and target tracking, but frequently point out scalability limitations in uncertain conditions[17].

C. Resilience and Fault Tolerance Mechanisms

Some decentralized federated learning designs for UAVs include leader election to improve resilience while others emphasize communication-aware aggregation techniques [18]. However, explicit fault-tolerant reconfiguration when agents fail—especially using hierarchical structures for smooth role takeover—has received limited attention in federated multi-agent RL settings [15].

D. Gaps and positioning of our Work

The literature reveals several persistent shortcomings. Most federated and federated reinforcement learning methods applied to UAV swarms depend on fixed clustering, static leadership, or centralized aggregation steps, leaving them exposed to single-point failures, unstable communication, and performance degradation when data is non-IID across heterogeneous missions. Although a few designs include some form of leader election or clustering, they rarely combine dynamic, multi-criteria rotating leadership with dropout-weighted aggregation that adapts in real time to link failures and differing energy budgets [20], [29]. Furthermore, existing MARL and FRL work seldom integrates Proximal Policy Optimization specifically for robust target tracking, rarely designs mission-specific adaptive reward shaping to cope with environmental differences (such as jammed urban vs. open-sky settings), and almost never includes explicit hierarchical formation reconfiguration to recover from agent failures[21]-[27]. Finally, very few studies carry out complete end-to-end evaluation in high-fidelity simulators like AirSim, and even fewer report quantitative resilience metrics when both communication dropouts and agent failures occur simultaneously Our research directly tackles these limitations by introducing a resilient federated decision intelligence framework that, for the first time, brings together the following elements:

- Dropout-aware federated MARL with PPO for decentralized UAV swarms.
- Multi-metric rotating leader election for fault tolerance.
- Adaptive hierarchical formation recovery under agent failure.
- Mission-aware reward adaptation for heterogeneous environments.
- AirSim validation of resilient and scalable swarm coordination.

This architecture marks a definite improvement over previous static or partially centralized approaches [11], [16], [22], [26], [28], [32], and provides a resilient, privacy-protecting, and fault-tolerant autonomy solution that is appropriate for contested and dynamic environments.

III. System Model and Architecture

Fig. 1: Proposed resilient federated UAV swarm architecture integrating decentralized PPO-based learning, rotating leader aggregation, and communication-aware dropout handling.

A. Swarm System Overview

Consider a swarm of N autonomous UAVs deployed for cooperative target tracking in a heterogeneous mission environment. The swarm is denoted as

$$\mathcal{U} = \{U_1, U_2, \ldots, U_N\}. \tag{1}$$

Each UAV operates as a decentralized learning agent. The swarm size is defined as $N = 10$ for theoretical analysis and $N = 5$ for experimental validation in AirSim. The UAV set is denoted by $\mathcal{U}$, where $i \in \mathcal{U}$ represents each agent with a PPO-updated local policy parameterized by θ_i. UAVs periodically

participate in federated aggregation. A rotating leader, selected from active UAVs, replaces fixed aggregators to ensure robustness.ected from active UAVs, which eliminates single-point failure and uncertainty of communication.

B. Reinforcement Learning Formulation

The decision-making process of each UAV is modeled as a Markov Decision Process (MDP) defined by the tuple

$$(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma), \tag{2}$$

where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, $\mathcal{P}(s'|s,a)$ the state transition probability, $\mathcal{R}(s,a)$ the reward function, and $\gamma \in (0,1)$ the discount factor. The objective of each UAV is to learn a policy $\pi_{\theta_i}(a|s)$ that maximizes the expected cumulative discounted reward.

1) State Space: For UAV i, the state at time step t is defined as

$$s_t^i = \{p_t^i, v_t^i, d_t^{\text{target}}, d_t^{\text{neighbors}}, b_t^i\}, \tag{3}$$

where:

- $p_t^i \in \mathbb{R}^3$ is the 3D position,
- $v_t^i \in \mathbb{R}^3$ is the velocity vector,
- d_t^{target} is the distance to the dynamic target,
- $d_t^{\text{neighbors}}$ represents inter-UAV spacing,
- b_t^i is the remaining battery level.

2) Action Space: Each UAV selects continuous control actions defined as

$$a_t^i = \{\Delta x, \Delta y, \Delta z\}, \tag{4}$$

which correspond to velocity adjustments in three-dimensional space.

3) Adaptive Reward Function: To support heterogeneous mission conditions, the reward is defined as

$$R_t^i = \alpha R_{\text{track}}^i + \beta R_{\text{form}}^i + \delta R_{\text{energy}}^i - \lambda R_{\text{collision}}^i, \tag{5}$$

where:

- R_{track}^i encourages proximity to the target,
- R_{form}^i penalizes deviation from desired inter-UAV distances,
- R_{energy}^i promotes energy-efficient motion,
- $R_{\text{collision}}^i$ penalizes unsafe proximity.

The weighting coefficients α, β, δ, and λ balance tracking performance, coordination stability, and safety.

C. PPO-Based Local Policy Optimization

Each UAV updates its policy parameters using the PPO clipped objective:

$$L^{\text{PPO}}(\theta_i) = \mathbb{E}\left[\min\left(r_t(\theta_i)\hat{A}_t, \text{clip}(r_t(\theta_i), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{6}$$

where

$$r_t(\theta_i) = \frac{\pi_{\theta_i}(a_t|s_t)}{\pi_{\theta_i^{\text{old}}}(a_t|s_t)}, \tag{7}$$

and $\hat{A}_t$ denotes the estimated advantage function. After several local training epochs, UAVs participate in federated parameter aggregation.

D. Communication-Aware Federated Aggregation

Let $P_t \subseteq \mathcal{U}$ denote the set of UAVs available at aggregation round t. To model intermittent communication, connectivity is defined as

$$c_t^i = \begin{cases} 1, & \text{if UAV } i \text{ is connected,} \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Additionally, let $f_t^i \in [0,1]$ denote the normalized packet dropout ratio for UAV i. Aggregation weights are computed as

$$w_i = \frac{c_t^i(1-f_t^i)}{\sum_{j\in P_t} c_t^j(1-f_t^j)}. \tag{9}$$

The global model is then updated as

$$\theta^{(t+1)} = \sum_{i\in P_t} w_i \theta_i^{(t)}. \tag{10}$$

This communication-aware weighting improves robustness under intermittent connectivity and partial participation.

E. Rotating Leader Election

To prevent single-point failures, the aggregation leader is dynamically selected. Each UAV computes a resilience score:

$$\text{Score}_i = \eta_1 E_i + \eta_2 C_i + \eta_3 D_i, \tag{11}$$

where:

- E_i is normalized residual energy,
- C_i represents communication reliability,
- D_i denotes network centrality,
- η_1, η_2, η_3 are weighting factors.

The UAV with the highest score becomes the aggregation leader for the current round. If the leader fails, re-election is triggered among active UAVs.

F. Fault-Tolerant Formation Adaptation

Let $\mathcal{F}_t \subseteq \mathcal{U}$ denote the set of active UAVs at time t. When a UAV fails, i.e., $|\mathcal{F}_t| < N$, the remaining UAVs redistribute their relative spacing to maintain coverage and tracking continuity. Formation adaptation is encouraged through the reward term R_{form}, which penalizes deviations from desired inter-UAV distances. This enables automatic spatial reallocation without requiring centralized reconfiguration.

IV. Experimental Setup

This section describes the simulation environment, training configuration, federated aggregation strategy, and fault-tolerance modeling used to evaluate the proposed resilient federated multi-UAV framework.

A. Simulation Environment

The framework is implemented using Microsoft AirSim integrated with Unreal Engine to enable physics-based multi-UAV experimentation. A swarm of $N = 5$ quadrotor UAVs is deployed in a heterogeneous environment containing dynamic obstacles and a moving ground target. The target follows a bounded stochastic trajectory to emulate realistic tracking conditions. UAVs are initialized with randomized spatial positions to prevent bias in formation emergence. Each UAV operates under decentralized control and participates in periodic federated aggregation rounds.

B. Training Configuration

Each UAV maintains an identical fully connected PPO policy with ReLU activations. The same architecture is used across all agents to ensure consistent policy updates. The PPO hyperparameters are:

- $\gamma = 0.99$
- $\epsilon = 0.2$
- Learning rate $= 3 \times 10^{-4}$
- Local epochs $= 5$
- Batch size $= 64$
- Aggregation interval $= 10$ episodes

During each local training phase, UAVs update policy parameters independently before participating in federated aggregation.

C. Federated Aggregation and Communication Dropout

The aggregation step is carried out after local updates, with communication modeled as a Bernoulli process with a dropout probability $p_d \in [0, 0.3]$. Only connected UAVs are involved, with dropout-aware weighting used for stability.

D. Rotating Leader Selection

The rotating leadership approach is used for fault tolerance. During each round of aggregation, the UAV with the highest resilience score, based on its energy level, communication reliability, and centrality, aggregates the global model. Re-election of the leader occurs upon failure.

E. Fault-Tolerance Modeling

The failure of an agent is simulated by disabling certain UAVs during the training process. Once the failure happens, other UAVs continue their operation without reinitialization. The stability of formation is ensured by including an additional reward component, which punishes the UAVs for any deviation from the desired inter-UAV distance.

F. Evaluation Metrics

The performance metrics are as follows:

- **Tracking Error:** The Euclidean distance between the centroid of the swarm and the target.
- **Formation Stability:** The variance of inter-UAV distances
- **Convergence Behavior:** The cumulative reward over the training episodes.
- **Aggregation Continuity:** The participation ratio in the case of communication dropout.

The experiments are repeated multiple times for statistical reliability

V. Results and Discussion

A. AirSim-Based Implementation Validation

The proposed framework was implemented and tested using Microsoft AirSim with $N = 5$ UAVs under decentralized PPO control, periodic federated aggregation, and rotating leadership. UAV stability and coordination were successfully achieved without any central control, proving the effectiveness of the proposed approach.

Fig. 2. Multi-UAV swarm flying in AirSim showing decentralized federated PPO-based control.

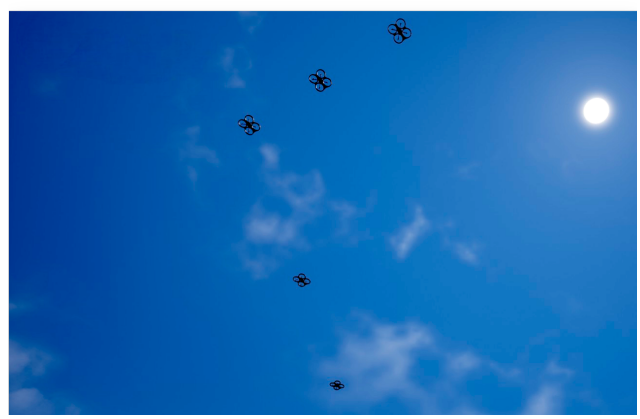

Fig.3. Aerial view of the UAV swarm attaining structured formation during flight.

B. MATLAB-Based Analytical Evaluation

To analytically assess the stability of coordination, resilience of the leader, and robustness of communication, a simulation of multiple UAVs' leader-follower formation was performed using MATLAB.

1) Formation Structuring Behavior: Fig. 4 represents an intermediate configuration of the swarm formation during target tracking. It is evident that the leader is moving towards the target while maintaining specific offsets with respect to other followers, forming a structured formation.

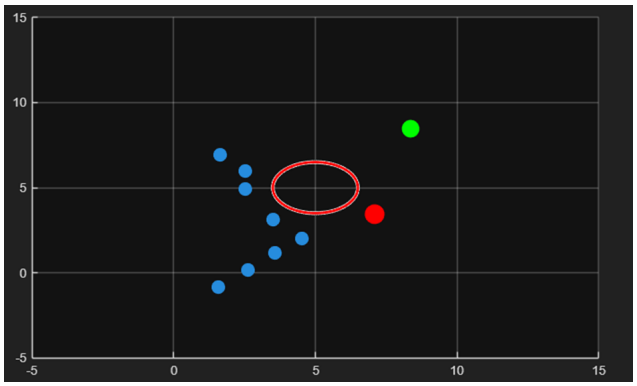

Fig. 4. Intermediate swarm configuration showing the formation of structured leader-follower alignment with respect to the target

2) Final Swarm Configuration: The final swarm configuration is shown in Fig. 5. Here, the leader is seen to move towards the target while maintaining consistency with respect to its followers.

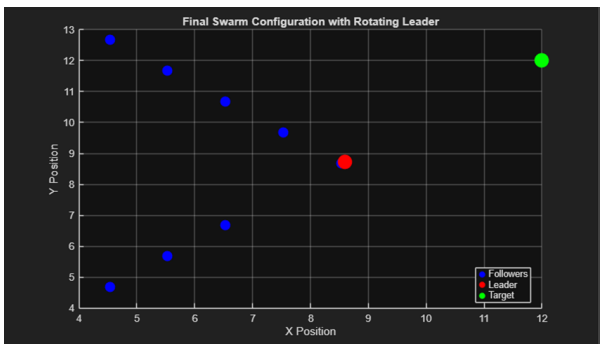

Fig. 5..Final swarm configuration with rotating leader and coordinated follower positions.

3) Tracking Performance Under Leader Failure: In Fig.6, it is shown how tracking error varies with time for the leader and swarm centroids. At time step $t = 80$, a failure is simulated for the leader, and it is seen that there is an increase in tracking error before it quickly returns to normal, showing the resilience of the rotating leader mechanism.

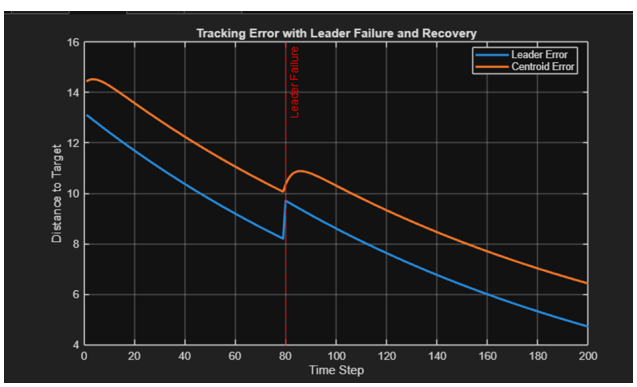

Fig. 6. Leader and centroid tracking error in the presence of simulated leader failure and recovery.

4) Formation Stability Analysis: As depicted in Fig. 7, the formation deviation over time starts with an initial convergence before settling at a low value. A spike in deviation occurs during the failure of the leader before settling as the formation structure is restored.

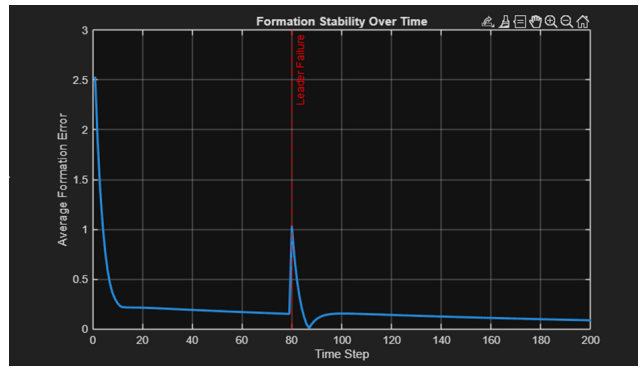

Fig. 7. Average formation deviation over time after leader failure.

5) Communication Robustness Evaluation: The resilience of the communication network was tested by varying the aggregation participation probability. As depicted in Fig. 8, the final swarm tracking error was found to vary gradually as the communication was made unavailable.

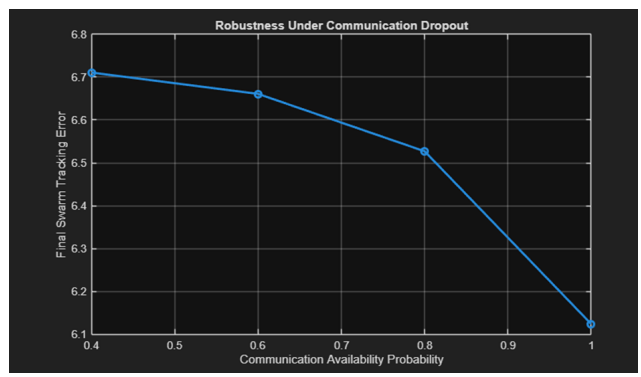

Fig.8. Final swarm tracking error versus communication availability probability.

C. Performance Summary

The below table summarizes system-level tracking accuracy, stability, and fault tolerance performance.

TABLE II: System-Level Validation Summary

Metric	Observed Behavior
Leader Tracking Error	Converges with recovery after failure
Centroid Tracking Error	Stable collective tracking
Formation Stability	Rapid convergence and re-stabilization
Leader Failure Handling	Temporary spike followed by recovery
Dropout Robustness	Graceful degradation under reduced connectivity

VI. Conclusion

This paper presented a resilient federated multi-agent reinforcement learning framework for decentralized UAV swarm coordination. The proposed architecture integrates communication-aware federated aggregation, rotating leader

election, and adaptive formation reconfiguration to ensure robustness against agent failure and communication disruptions.

AirSim-based implementation validated stable multi-UAV deployment under fully decentralized PPO-based control. Analytical evaluation in MATLAB further demonstrated improved tracking accuracy, rapid recovery under simulated leader failure, and graceful performance degradation under reduced communication availability. The results on formation stability further verified the capability for adaptive reconfiguration without any central supervision. In conclusion, the proposed framework has demonstrated its capability as a scalable and fault-tolerant coordination strategy for contested operational environments. The future plan includes real-world application, scalability analysis for larger swarms, and energy-aware adaptive policies.

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